# THIRD SEMESTER B.A./B.Sc. DEGREE EXAMINATION
## NOVEMBER 2019

(CUCBCSS—UG)

## A 11—BASIC NUMERICAL SKILLS

(2014 Admissions)

**Time : Three Hours** **Maximum : 80 Marks**

### Part I

*Answer all questions.*
*Each question carries 1 mark.*

Choose the correct answer :

1. If the simple Interest for 3 years @ 10 % is Rs. 3,000, the principal is :

   (a) 20,000. (b) 3,000.

   (c) 12,000. (d) 10,000.

2. The matrix obtained by interchanging the rows and columns is known as :

   (a) Transpose of matrix. (b) Determinant.

   (c) Inverse of matrix. (d) Unity matrix.

3. Frequency distribution is :

   (a) Total frequency. (b) A variable.

   (c) A table. (d) Class intervals.

4. The difference between two adjacent numbers in the A.P is called :

   (a) Common difference. (b) Progression.

   (c) Common ratio. (d) Ratio.

5. The weight used in Paasche's formula belongs to :

   (a) Base period. (b) Current period.

   (c) Any arbitrary chosen period. (d) None.

Fill in the blanks :

6. A time series is a set of values arranged in ——— order.
7. $A^{-1}$ means ———.
8. Index numbers shows ——— changes rather than absolute changes.
9. When A = {a, b}, its power set has ——— elements.
10. Sequence of Numbers is called ———.

(10 × 1 = 10 marks)

## Part II

*Answer any* **eight** *questions.*
*Each question carries* **2** *marks.*

11. What is secular trend ?

12. Solve $\frac{15}{x+4} = \frac{10}{x-3}$.

13. Find the median for the following values :

    4, 45, 60, 20, 83, 19, 26, 11, 27, 12, 52.

14. Rs. 10,000 for 3 years @ 15 % p.a. Calculate the simple interest and amount for the investment.

15. Find the $15^{th}$ and $20^{th}$ terms in the series 2, 8, 14, 20....

16. Find Range :

| | | | | | | |
|---|---|---|---|---|---|---|
| Wage | : | 10 | 15 | 18 | 20 | 25 |
| No of Employer | : | 3 | 5 | 12 | 8 | 6 |

17. Which term of the A.P 49, 44, 39, ..... 9 ?

18. Calculate determinant of $A = \begin{bmatrix} 2 & 7 \\ 5 & 3 \end{bmatrix}$.

19. If the population of a village now is 62,000 and the rate of growth of population per year in 30 per 1000, find the expected population at the end of $4^{th}$ year :

20. Find the sum and product of the roots of the equation $x^2 + 5x + 9 = 0$.

(8 × 2 = 16 marks)

## Part III

*Answer any* **six** *questions.*
*Each question carries 4 marks.*

21. What is Graph ? What are the uses of graphs ?

22. Solve $xy + x + y = 27, \frac{1}{x} + \frac{1}{y} = \frac{1}{2}$.

23. Find the sum to $n$ terms of the series 6 + 66 + 666 + ..... $n$ terms.

24. Calculate the total interest on Rs. 500 for 73 days, Rs. 720 for 14 weeks and on Rs. 900 for 3 months, all at 6 % per annum.

25. Calculate determinant of the matrix, $A = \begin{bmatrix} 2 & 7 & 3 \\ 6 & 4 & 8 \\ 1 & 2 & 5 \end{bmatrix}$.

26. If A = {1, 2, 3}, B = {3, 4, 5), C = {1, 3, 5}, prove that $A - (B \cup C) = (A - B) \cap (A - C)$.

27. Calculate simple index number by average relative method.

| Items | : | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Price in base year | : | 5 | 10 | 15 | 20 | 8 |
| Price in current year | : | 7 | 12 | 25 | 18 | 9 |

28. What is Statistics and explain the features ?

**(6 × 4 = 24 marks)**

## Part IV (Essay Questions)

*Answer any* **two** *questions.*
*Each question carries* 15 *marks.*

29. Explain the different stages in a statistical enquiry

30. If $A = \begin{bmatrix} 1 & 5 & 2 \\ 3 & -2 & 4 \\ 6 & 2 & 1 \end{bmatrix}$, find the inverse of A.

31. From the following indices calculate the weighted cost of living index numbers for the year 2014, 2015 and 2016 :

| *Items* | *Weight* | *Indices* | | | |
|---|---|---|---|---|---|
| | | 2013 | 2014 | 2015 | 2016 |
| Food | 50 | 100 | 110 | 115 | 120 |
| Clothing | 30 | 100 | 100 | 110 | 105 |
| Housing | 20 | 100 | 95 | 120 | 104 |
| Fuel | 15 | 10 | 105 | 108 | 106 |
| Miscellaneous | 5 | 100 | 102 | 112 | 110 |

**(2 × 15 = 30 marks)**